ATHIF FADHEEL ATHARAHMAN

Robotics & Systems Engineer | Industrial Automation (SCADA / PLC) | Data Engineering

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PROFILE

Electrical Engineering graduate who builds systems end to end: UAV flight software and telemetry, PLC and SCADA integration for industrial operations, and the data pipelines and dashboards that turn field signals into decisions. Team Responsible and lead programmer for Telkom University's fixed-wing UAV team at TEKNOFEST (Türkiye), KRTI National Top 8 finisher, and former SCADA & Automation intern at PT PLN UIP2B JAMALI. Author of 3 conference papers spanning UAV communication, computer vision, and AI, and holder of 1 granted patent.

TECHNICAL SKILLS

Robotics & UAV: ROS, Gazebo, ArduPilot, Lua scripting, MAVLink, Mission Planner, flight dynamics, LoRa & 4G-LTE telemetry
Embedded & Vision: C/C++, Python, Raspberry Pi, OpenCV, MediaPipe, YOLO, TensorFlow / TFLite, facial recognition
Industrial Automation: SCADA, PLC / Ladder Logic (CX-Programmer), Factory I/O, OPC, ICCP / TASE.2 (IEC 60870-6), FAT
Data & Backend: SQL, Pandas, NumPy, ETL, Scrapy, Playwright, Celery, FastAPI, Node.js, REST APIs, Grafana, Jupyter, Excel / VBA
AI & LLM: LangGraph agentic workflows, prompt engineering, model evaluation & QA, Vertex AI, OCR, edge AI, data annotation
Databases & Cloud: Oracle, SQL Server, MySQL, PostgreSQL, Redis, Supabase, GCP (Cloud Run), AWS EC2, Azure, Firebase, Docker
Systems & Web: Linux, NAS (RAID 0/1), TCP/IP, VPN (Tailscale), Git, TypeScript, React / React Native, Next.js, Three.js, Vercel
Engineering & Professional: MATLAB / Simulink, SolidWorks, Ansys, technical project management, cross-functional leadership

PROFESSIONAL EXPERIENCE

SCADA & Automation Division Intern **Jun 2025 - Aug 2025**
PT PLN (Persero) UIP2B JAMALI | Depok, Indonesia

- Automated downtime and reliability reporting with Python scripts that parse, clean, and aggregate server and RTU logs.
- Queried operational data in Oracle, SQL Server, and MySQL; built Grafana dashboards for trends, alarms, and system health.
- Prototyped an ICCP / TASE.2 gateway in C on the libiec61850 MMS stack; built from source and debugged interoperability.
- Ran FAT on 48V DC rectifier and Ni-Cd battery systems; documented configurations, results, root causes, and fixes.
- Administered NAS / RAID storage, static IP, user access, and secure connectivity; diagnosed hardware and network faults.

ROBOTICS & UAV LEADERSHIP

Team Responsible & Lead Programmer **2025 - 2026**
Team Avalerion APTRG, Fixed Wing Category | TEKNOFEST, Türkiye, Malatya

- Led flight software and autonomous mission planning for Telkom University's fixed-wing UAV at the TUBITAK competition.
- Coordinated technical milestones, hardware integration, and deliverables to the international committee.

Technical Supervisor & Lead Programmer **2024 - 2025**
Team Frigate APTRG, Racing Plane Division | KRTI, Indonesia, Padang

- Led technical coordination across a 10+ member multidisciplinary team and achieved a National Top 8 finish.
- Maintained autopilot scripts, ran telemetry integration tests, and resolved hardware failures in the field.

Data Systems Programmer & Telemetry **2023 - 2024**
Team Strix APTRG, Long Endurance Low Altitude (LELA) Division | KRTI, Indonesia, DI Yogyakarta

- Built RF and 4G-LTE telemetry pipelines with store-and-forward cloud messaging for long-endurance autonomous flight.

Vice Captain & Deputy External Affairs Coordinator **2024 - 2025**
APTRG Laboratory | Bandung, Indonesia

- Managed schedules and technical documentation; handled sponsor, alumni, and university administration relations.

PUBLICATIONS & PATENT

- **ICCTEIE 2025 (IEEE Xplore):** *Performance & Reliability Analysis of 4G-Based Flight Controllers in UAV Control.*
- **EECSI 2026:** *Adaptive Blink Detection Using MediaPipe FaceMesh and Sigmoid-Based Eye-Closure Scoring Under Low-Light Conditions.*
- **ICVEE 2026 (accepted):** *The Efficacy of Large Language Model Tools in Solving Electrical Circuit Problems.*
- **Granted patent:** Drowsiness Detection Alert, edge computer-vision fatigue alert on Raspberry Pi with infrared imaging at 0 lux.

SELECTED PROJECTS

- **PLC Warehouse Traffic Control (CX-Programmer, Factory I/O, OPC):** sorting variable-height cargo with conveyor interlocks.
- **NFCC Digital Twin (React, Three.js, Web Workers):** command-center twin simulating 5,000 sensor nodes at 60 fps.
- **AIGents (LangGraph, FastAPI, SQLite, Docker):** state-machine AI agent that loops until a high-confidence solution.
- **Price-Intelligence Scraper (Scrapy, Playwright, Celery, PostgreSQL, Docker):** async pipeline, 1,100+ products tracked.
- **URLSiap (Next.js, Supabase, Zod):** SEO and performance audit tool with SSRF hardening and persistent job tracking.
- **Otentikasi (React Native, Firebase, Vertex AI, TensorFlow):** mobile user verification via facial recognition.

EDUCATION

Bachelor of Electrical Engineering, Cum Laude (GPA 3.53 / 4.00) | Telkom University, Bandung **2022 - 2026**
Coursework: Embedded Systems & IoT, Machine Learning & AI, Computer Networks, Database Systems, Data Structures

CERTIFICATIONS & LANGUAGES

- **Oracle AI Database Certified Foundations Associate** — *Oracle University | Aug 2026*
- **Databases for Developers: Foundations** (100%) & Next Level (97%) — *Oracle Dev Gym*
- **Industrial SCADA & Power Systems Automation**, practical training — *PT PLN (Persero) UIP2B JAMALI | 2025*
- **English Proficiency:** EPrT 547 (CEFR B2), ECCT 3.75; Indonesian (native) — *Telkom University*